modified to reflect that of the next node in the path. This way, each intermediary can only discover (by using the necessary tools) the destination of the packet, not its source. Only at the final node are all the packets unencrypted and put together and sent to the destination.

Checking if TOR is doing its job

The TOR application is bundled with a few other tools that make it easier to use it. Besides the core TOR program, the bundle consists of a proxy server (Privoxy), a GUI to control TOR (Vidalia), and a Firefox plugin (Torbutton) which allows Firefox to work with TOR. Running TOR is just the first step: to use it, the browser and every communication tool needs to be configured to send data through it. This can be done by changing the network/connection settings to use a proxy server. This needs to be set to IP 127.0.0.1 and port 8118.

Once this is done, the user should test if the network is doing what it is supposed to do by visiting any site that offers information about the external IP of the user (for example, ip-address.domaintools.com). The IP reported should be different from the value that is visible by running the “ipconfig” command at a command prompt. In fact, it will be the same as one of the intermediaries in the path. Alternatively, visiting http://torcheck.xenobite.eu /index.php will give you specific information about how the TOR installation is performing. The Vidalia control panel can be used to monitor and configure TOR.

The View the Network Button opens a window listing the nodes in TOR in the left frame. The active paths available for use by the user are listed in the middle frame, and clicking on any of these paths will reveal the details of the nodes in them in the right path. The world map will show the location of the nodes. To see the network in action, try accessing any Web site from the browser and switch to the Network map window—and watch the middle frame.

The Bandwidth graph button shows bandwidth statistics in real-time.